VACATION-RENTAL MARKETPLACE — PRODUCTION SCALE ARCHITECTURE

Scaling Strategy: Frontend · Backend · Storage · Search · Deployment (Airbnb 100M+ Req/day)

TIER 1: GLOBAL CLIENTS & EDGE ACCELERATION (CLOUDFLARE + CLOUDFRONT 310+ POPS)

- Next.js / React Web
  SSR/ISR Edge Hydration
- Mobile iOS & Android
  React Native / Native
- Global Anycast Edge CDN · DDoS Shield (L3/L4/L7)
  On-Demand ISR Purging · Dynamic WebP/AVIF Transcoding

TIER 2: API GATEWAY, RATE LIMITING & BACKEND-FOR-FRONTEND (BFF)

- Envoy Proxy & WAF
  mTLS · Circuit Breaking
- Token Bucket Rate Limiter
  IP & User Key Throttling
- JWT & OAuth2 Auth
  Stateless Signature Check
- GraphQL Federation / BFF
  Apollo Subgraph Router

TIER 3: CORE MICROSERVICES (KUBERNETES EKS AUTO-SCALING)

- Listing Service
  Metadata & Rules
- Search & Discovery
  Geo-Hash Filters
- Booking Saga Engine
  Redlock Concurrency
- Dynamic Pricing ML
  Seasonality & Yield
- Payment & Trust
  Stripe/Razorpay Hooks

TIER 4: STORAGE, REAL-TIME SEARCH & EVENT BUS

- PostgreSQL Aurora Shards
  Sharded by listing_id
  Strict Serializable ACID
- Redis Availability Cluster
  365-Day Calendar Bitmaps
  Sub-ms BITOP Range
- Elasticsearch Cluster
  Spatial geo_point Candolim
  Facet Aggregations (P99 < 30ms)
- Apache Kafka Event Spine
  Topics: booking.held, updated
  Event Sourcing & Replay

TIER 5: ASSET STORAGE, ANALYTICS & MULTI-REGION DEPLOYMENT

- Cloudflare R2 / AWS S3
  Direct Presigned Uploads
  Sharp Image Pipeline
- Snowflake / ClickHouse OLAP
  Data Lakehouse for ML Yield
  Host Metrics & Booking Analytics
- Multi-Region Kubernetes (EKS)
  Active-Active (US/EU/Asia)
  ArgoCD GitOps & OpenTelemetry